
Alexandria University
Faculty of Engineering
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AEGIS

AI-Enhanced Guardian & Intelligent Security System

1. Introduction

AEGIS (AI-Enhanced Guardian & Intelligent Security System) is a graduation project that delivers an AI-augmented Security Operations Center (SOC) platform combining industry-grade SIEM/SOAR infrastructure with custom-built artificial intelligence, decision logic, and analyst tooling.

1.1 SIEM & SOAR Strategy

Rather than building SIEM and SOAR engines entirely from scratch, the platform is built on a selected third-party SIEM/SOAR solution that supports deep customization and pipeline integration. This gives the project a stable, battle-tested foundation for log correlation, alert lifecycle management, and playbook execution.

What the team builds on top of that foundation is where the project's engineering value lives:

- Choose the right SIEM and SOAR solutions and build the infrastructure on which we will test it.
- Custom AI models for threat detection, severity classification, and vulnerability assessment, integrated directly into the SIEM's detection pipeline.
- A rule-based Decision Engine that fuses rule-based and AI-based verdicts deterministically.
- A cloud-hosted infrastructure, containerized and orchestrated with Docker/Kubernetes, that reliably runs and scales all platform services.
- A fully custom-built SOC dashboard/portal

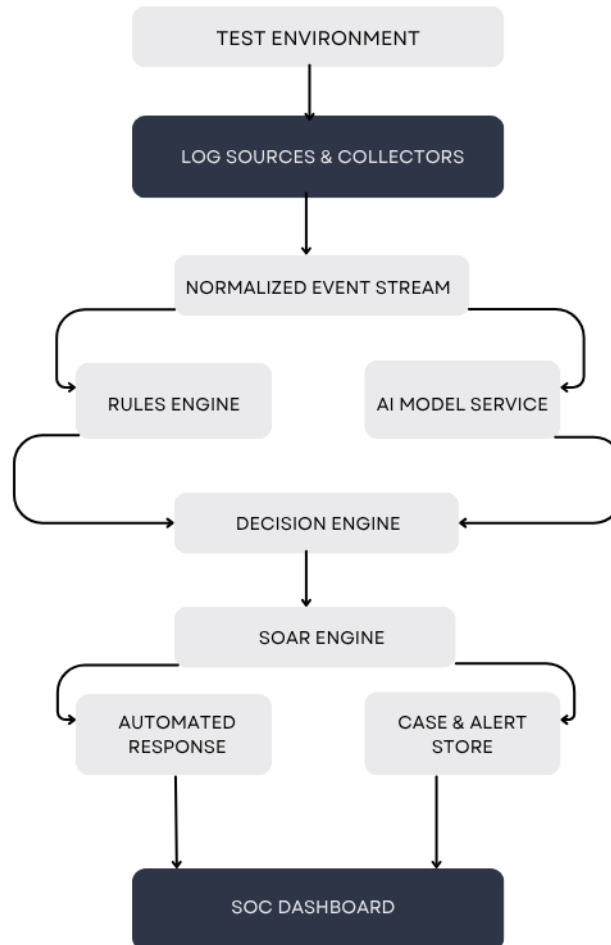
2. Standards & Threat Frameworks

Attack scenarios, detection rules, and AI model training are all grounded in recognized industry frameworks, ensuring the platform's threat coverage and terminology are defensible and audit-ready:

- **MITRE ATT&CK:** used to model attacker techniques and to map correlation rules and AI detections to known tactics/techniques.
- **OWASP Top 10:** used to scope web/API vulnerability classes covered by the vulnerability assessment model and Red Team testing.
- **ISO/IEC 27001:2022:** used as the governance baseline for information security controls, asset classification, and risk treatment.

3. High-Level Architecture

The diagram below illustrates the end-to-end flow of the platform, from raw log ingestion through AI-assisted decision making to automated response and SOC visibility.



4. Ideation

The following outlines the proposed end-to-end detection-and-response workflow, from log ingestion through SOC analyst review:

- **Foundation and log sources:** Third-party SIEM and SOAR solutions will serve as the backbone of the platform. The system will collect logs from multiple sources, including endpoints, networks, cloud environments, APIs, and, optionally, a test website.
- **Aggregation and parallel processing:** All collected logs are aggregated and normalized into a unified format, then routed through two parallel processing paths:
 - **Path 1 — Condition Engine:** rule-based detection using Sigma rules mapped to MITRE ATT&CK. For example, if MITRE ATT&CK technique T1133 (External Remote Services) is detected, the engine automatically raises an alert.
 - **Path 2 — AI Model Service:** a set of custom-built AI models, including a threat detector, a severity classifier, and a vulnerability assessment model.
- **Decision Engine:** Once both paths complete their analysis, the results are forwarded to a Decision Engine, which produces a single, final severity verdict. The Decision Engine acts as a business logic layer that applies rule-based fusion — a fully deterministic and explainable approach. For example: if a rule match exists and the AI confidence exceeds 0.8, the severity is classified as Critical.
- **SOAR processing:** The verdict is then passed to the SOAR platform, which performs:
 - **Alert enrichment:** for example, checking whether an IP address is malicious via a VirusTotal lookup.
 - **Playbook matching:** identifying and triggering the appropriate incident response playbook.
- **Response action:** The SOAR platform then takes one of two response actions:
 - **Automated response:** such as blocking a malicious IP address.
 - **Case storage:** logging and storing the alert for further review.
- **SOC review:** Finally, all processed alerts and cases are forwarded to the custom-built SOC dashboard, where a SOC analyst reviews the data and makes the final decision.

5. Team Responsibilities

5.1 Security Team

- Select the appropriate third-party SIEM solution that supports customization and integration with our pipeline.
- Handle log collection and normalization — ingesting raw logs from all system components and mapping them into a unified schema.
- Write correlation rules mapped to the MITRE ATT&CK framework.
- Select the appropriate third-party SOAR solution.
- Design incident response workflows and automate playbook execution.

5.2 DevOps Team

Host the infrastructure on a cloud provider environment where:

- Cloud resources, databases, and networking will be managed and monitored.
- Collected information and system data will be stored and managed.
- Kubernetes and Docker images for containerized services will be managed.
- Infrastructure availability, security, and scalability will be ensured.

5.3 Full-Stack Team

- The Full-Stack Team is responsible for designing and developing the SOC portal, which serves as the main interface through which SOC analysts interact with the system.
- The platform will include a secure login system, a main dashboard providing an overview of the security environment
- Specialized dashboards for supported system domains such as Network, APIs, Cloud, Endpoint Devices, and Website monitoring (some of which may be optional).
- The portal will provide alert management capabilities, allowing analysts to view, search, filter, and investigate security alerts.
- Detailed alert views will present relevant information received from the SIEM, AI models, Decision Engine, and SOAR platform.

5.4 AI Team

- Develop an AI model for threat detection.
- Develop an AI model for vulnerability classification into severity levels: Critical, High, Medium, and Low.
- Develop an AI model for vulnerability assessment.
- Train all models based on industry frameworks and standards: OWASP Top 10, ISO 27001, and MITRE ATT&CK.

6. Timeline

First Month — September to October

- Conduct cost optimization analysis for the infrastructure.
- Produce a low-level design document for the full system.
- Distribute roles and responsibilities across all teams.
- Define and finalize the full project timeline for the year.
- Investigate whether custom AI models can be integrated directly within the chosen third-party SIEM solution.

The full year timeline will be finalized and shared by the end of September.

7. About the Team

The project is developed by a multidisciplinary team of eight engineers spanning cybersecurity, machine learning, full stack development, DevOps, and penetration testing.

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